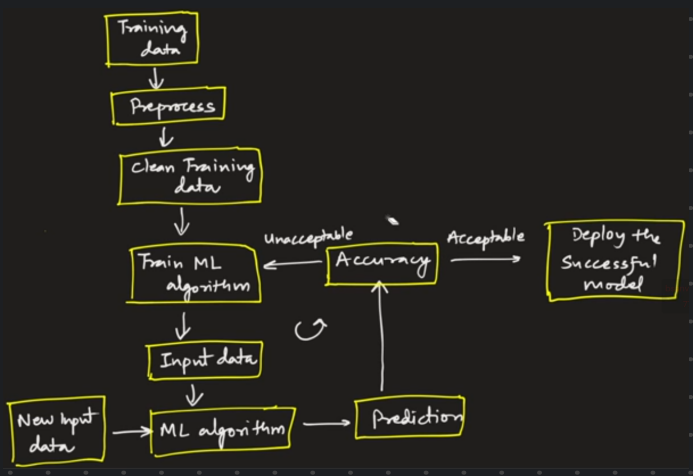
॥ श्रीगणेशाय नमः ॥

Machine Learning

ML is a subset of AI that focuses on development of algorithms that allows computers to learn from and make predictions or decisions based on data.
Instead of explicitly programming rules, ML algos identify patterns in data and improve performance over time.
An ML model is similar to a mathematical function. It takes input data and predicts and output.

ML
- supervised : model is trained on a labelled dataset
- unsupervised : model is trained on an unlabelled dataset, trying to identify hidden patterns.
- Reinforcement : model reacts with env. learns by rewards and penalties.

How does ML work?



ML problems

Regression: Predict continuous values. E.g. stock price prediction.
Classification: Classify the data points into classes based on features.

- Clustering: Forming clusters of data points.
- Discriminant analysis: Projects the data to lower-dimensional space to maximize class separability.
- Support vector machine: Finding the best hyperplane that best separates data into classes.
- Decision trees: Tree like model of decisions and their possible consequences.
- Ensembling: Build multiple ML models and merge the results.

Regression

Regression is a statistical method used for predicting continuous outcomes (dependent variable) based on one/more input features (independent variables).
Purpose: To identify the relationship between variables, understand patterns, and make future predictions.
E.g. Predict someone's weight based on their height, predict some price based on features like size, location, # of rooms, etc.

Goal: Given a training set comprising of N observations $\{X_n\}$, n=1, 2, .., N, together with corresponding target values $\{t_n\}$, the goal is to predict the value of t for a new value of X.

$X_1 = \{5'5, \text{male India}\}$ $t_1 = 50kg$
$\vdots$
$X_n = \{$ $\}$ $t_n = 60kg$

new value $X = \{\cdots\}$, $t = ?$

Common regression algos:
- Simple linear regression: relationship betn 2 variables.
- Multiple linear regression: relationship betn 1 dependent and multiple independent variables.
- Polynomial regression: extension of linear regression by adding polynomial terms.
- Ridge and lasso regression: regularization techniques.
- Logistic regression: classification technique, not regression.
- Multivariate adaptive regression splines.

Simple linear regression

E.g.

Study hrs	score
2	12
4	16
8	28
16	62

What'll be the exam score for a student who studied 5 hrs?

Equation of the regression line: $y = mx + c$ (slope: m; y-intercept: c; output/dependent: y; input var / independent / predictor: x)

Cost function/error function/loss function:
To measure the accuracy of a regression line, we use a cost function.
The most common cost function for regression is mean squared error (MSE). It calculates the average of the squared differences between the actual and predicted values.

Sum of squared errors:

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y_i})^2$$

Mean squared error:

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y_i})^2$$

Actual output	predicted output
5	6
4	4
2	1

$SSE = (5-6)^2 + (4-4)^2 + (2-1)^2 = 2$

$MSE = \frac{2}{3}$

Least squares method (to minimize the cost function)

$y = mx + c$ (m, c: hyperparameters)

For i^{th} data point, $y_i = mx_i + c$

The error (residual) for the ith data point:

$$Error_i = y_i - \hat{y_i} = (y_i - (mx_i + c))^2$$

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y_i})^2$$

$$SSE = \sum_{i=1}^{n} [(y_i - (mx_i + c))^2]$$

Partial derivative w.r.t m and c:

$$\frac{\partial (SSE)}{\partial m} = \frac{\partial}{\partial m} \left(\sum_{i=1}^{n} (y_i - mx_i - c)^2 \right) = \sum_{i=1}^{n} (-x_i)(2)(y_i - mx_i - c)$$

$$= -2 \sum_{i=1}^{n} x_i (y_i - mx_i - c)$$

$$\frac{\partial (SSE)}{\partial m} = 0 \Rightarrow \sum_{i=1}^{n} x_i y_i = m \sum_{i=1}^{n} x_i^2 + c \sum_{i=1}^{n} x_i \quad \text{——①}$$

$$\frac{\partial (SSE)}{\partial c} = \frac{\partial}{\partial c} \left(\sum_{i=1}^{n} (y_i - mx_i - c)^2 \right) = \sum_{i=1}^{n} (-1)(2)(y_i - mx_i - c)$$

$$\frac{\partial (SSE)}{\partial c} = 0 \Rightarrow \sum_{i=1}^{n} (y_i - mx_i - c) = 0 \Rightarrow \sum_{i=1}^{n} y_i = m \sum_{i=1}^{n} x_i - cN \quad \text{——②}$$

From ②, $c = \frac{\sum y_i - m \sum x_i}{N}$. Substituting in ①:

$$\sum x_i y_i = m \sum x_i^2 + \frac{\sum y_i - m \sum x_i}{N} \sum x_i$$

$$\Rightarrow m = \frac{\sum x_i \sum y_i - N \sum x_i y_i}{(\sum x_i)^2 - N \sum x_i^2}$$

$$m = \frac{N \sum x_i y_i - \sum x_i \sum y_i}{N \sum x_i^2 - (\sum x_i)^2} \qquad c = \frac{\sum y_i - m \sum x_i}{N}$$

Q. Consider a dataset of points:
Find the best fit line y = mx+c using the least squares method.

x	y
1	2
2	3
3	5
4	7
5	8

→

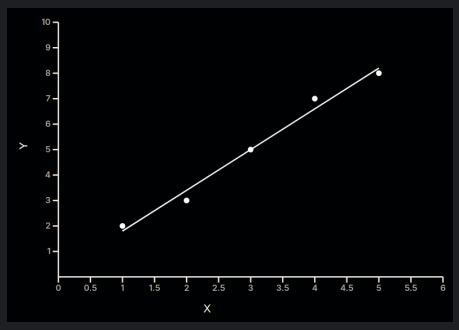
x_i	y_i	x_iy_i	x_i^2
1	2	2	1
2	3	6	4
3	5	15	9
4	7	28	16
5	8	40	25
15	25	91	55

$$m = \frac{N\sum x_iy_i - \sum x_i \sum y_i}{N\sum x_i^2 - (\sum x_i)^2}$$

$$m = \frac{5(91) - 15 \cdot 25}{5 \cdot 55 - (15)^2} = 1.6$$

$$c = \frac{\sum y_i - m\sum x_i}{N} = \frac{25 - (1.6)(15)}{5} = 0.2$$

$$y = 1.6x + 0.2$$



Q. Calculate the mean-squared error for the line on the given dataset.

→ $\hat{y} = 1.6x + 0.2$

x	y	$\hat{y}$	$\lvert y-\hat{y}\rvert$	$(y-\hat{y})^2$
1	2	1.8	0.2	0.04
2	3	3.4	0.4	0.16
3	5	5	0	0
4	7	6.6	0.4	0.16
5	8	8.2	0.2	0.04
				0.4 ← SSE

$$MSE = \frac{0.4}{5} = 0.08$$

Q. Consider a dataset of points:
Find the best fit line y = mx+c using the least squares method.

Study hrs	score
2	12
4	16
8	28
16	62

→

x	y	xy	x^2
2	12	24	4
4	16	64	16
8	28	224	64
16	62	992	256
30	118	1304	340

$m = 3.64$
$c = 2.17$

$$m = \frac{N\sum x_iy_i - \sum x_i \sum y_i}{N\sum x_i^2 - (\sum x_i)^2} = \frac{4(1304) - 30 \cdot 118}{4(340) - (30)^2} = \frac{1676}{460} = 3.6435$$

$$c = \frac{\sum y_i - m\sum x_i}{N} = \frac{118 - (3.6435)(30)}{4} = 2.17375$$

Gradient descent method (to minimize the cost function)

There won't always be a closed-form (→ direct formula) soln for the hyperparameters (it's only there for the least squares method). More often than not, we'll use GD for optimizaⁿ.

Predicted value: $\hat{y}_i = mx_i + c$

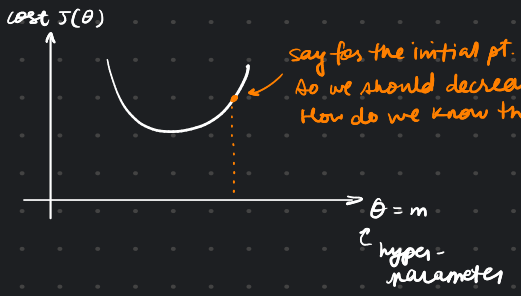
$$SSE(m,c) = \sum_{i=1}^{N} (y_i - (mx_i + c))^2$$

Step 1: Take an initial value for the hyperparameters. ← $m, c = 0$ (or some random value)

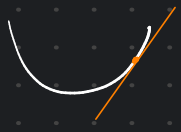
Step 2: Iteratively adjust the parameters in the direction that reduces cost. ← assumption: cost is a convex funcⁿ

$$m_{new} = m_{old} - \alpha \cdot \frac{\partial MSE}{\partial m}$$

$$c_{new} = c_{old} - \alpha \cdot \frac{\partial MSE}{\partial c}$$



say for the initial pt. we are here. So we should decrease m. How do we know that?



calculate gradient at that point. It is +ve (meaning the funcⁿ is increasing). So we need to decrease the hyperparameter.

If gradient at m_0 is +ve → decrease m_0

-ve → increase m_0

we want a slope 0.



value of cost funcⁿ that we'd prefer.

gradient of $J(\theta)$ at $\theta = \theta_0$ $= \left.\frac{\partial J(\theta)}{\partial \theta}\right|_{\theta=\theta_0}$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - (mx_i + c))^2$$

$$\frac{\partial MSE}{\partial m} = -\frac{2}{N}\sum_{i=1}^{N} x_i (y_i - (mx_i + c))$$

$$\frac{\partial MSE}{\partial c} = -\frac{2}{N}\sum_{i=1}^{N} (y_i - (mx_i + c))$$

Consider the following dataset with five observations:

X	Y	x^2	xy	
2	1	4	2	
3	2	9	6	
5	4	25	20	
7	5	49	35	
9	6	81	54	
$\Sigma x = 26$	18	168	117	

$$m = \frac{N\Sigma xy - \Sigma x \Sigma y}{N\Sigma x^2 - (\Sigma x)^2} = \frac{5(117) - 26 \cdot 18}{5(168) - 26^2} = 0.713$$

$$c = \frac{\Sigma y - m\Sigma x_i}{N} = \frac{18 - (0.713)(26)}{5} = -0.11$$

A simple linear regression model $y = mx + c$ is fitted to this data using the least squares method.

Which of the following statements are **true**?

- **A.** The slope m of the regression line is approximately 0.6. ✗
- **B.** The intercept c of the regression line is approximately -0.2. ✗
- **C.** The cost function used in the linear regression is defined as $J(m,c) = \frac{1}{2n}\sum_{i=1}^{n}(y_i - (mx_i + c))^2$. ✗
- **D.** Gradient descent method can be used to minimize the cost function in linear regression, but it may converge slowly if the learning rate is too low. ✓

Error metrics

In simple linear regression, the goal is to find the best fitting line that predicts the dependent variable Y from the independent variable X.
To evaluate the performance of the regression model, we use error metrics to quantify how well the model's predictions match the actual data.

Mean absolute error

- Average of the absolute differences between the actual and the predicted values.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

Sum square error

$$SSE = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Mean squared error

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 = \frac{SSE}{N}$$

Root mean squared error

$$RMSE = \sqrt{MSE} = \sqrt{\frac{\sum(y_i - \hat{y}_i)^2}{N}}$$

Mean absolute percentage error

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100$$

R-squared (coefficient of determination)

- Measures the proportion of variance in the dependent variable Y that is predictable from the independent variable X.

$R^2 \in [0, 1]$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}$$

← variance of y captured by x. (RSS: residual sum of squares)

← variance of y (TSS: total sum of squares)

worst: $\hat{y}_i = \bar{y} \Rightarrow R^2 = 1 - 1 = 0$

Best: $\hat{y}_i = y_i \Rightarrow R^2 = 1 - 0 = 1$

Adjusted r-squared

- Adjusts the R^2 value based on the # of predictors in the model.

$$R^2_{adj} = 1 - \frac{SS_{residual}/(N-K-1)}{SS_{total}/(N-1)} = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{(N-K-1)} \cdot \frac{(N-1)}{\sum_{i=1}^{N}(y_i - \bar{y})^2}$$

$N \rightarrow$ # of data points

$K \rightarrow$ # of predictors (1 for simple linear regression)

$K \uparrow \; R^2_{adj} \downarrow$

e.g. actual values: [3, 4, 5, 6], predicted values: [2.5, 4.2, 4.8, 6.3]

$MAE = \frac{1}{N}(0.5 + 0.2 + 0.2 + 0.3) = \frac{1.2}{4} = 0.3$

$SSE = (0.5^2 + 0.2^2 + 0.2^2 + 0.3^2) = 0.42$

$MSE = \frac{0.42}{4} = 0.105$

$RMSE = 0.324$

$MAPE = \frac{1}{4}\left(\frac{0.5}{3} + \frac{0.2}{4} + \frac{0.2}{5} + \frac{0.3}{6}\right) \times 100 = \frac{1}{4}\left(\frac{10 + 3 + 2.4 + 3}{60}\right) \cdot 100 = 18.4 \times \frac{25}{60} = 7.66\%$

$R^2 = 1 - \frac{0.42}{(1.5^2 + 0.5^2 + 0.5^2 + 1.5^2)} = 1 - \frac{0.42}{5} = 0.916$

$R^2_{adj} = 1 - \frac{0.42}{(4-1-1)} \cdot \frac{(4-1)}{5} = 1 - \frac{0.42}{5} \cdot \frac{3}{2} = 0.874$

(assuming model is simple lin. regr^n)

Q. Given the actual values, y_actual = [3, 5, 7, 9] and the predicted values, y_pred = [2.8, 4.9, 6.8, 9.1], which of the following statements are correct?

a. MAE = 0.15 b. SSE = 0.06 c. MSE = 0.015 d. R^2 = 0.995

$MAE = \frac{1}{N}\sum|y - \hat{y}| = \frac{1}{4}(0.2 + 0.1 + 0.2 + 0.1) = 0.15$

$SSE = \sum(y - \hat{y})^2 = (0.2^2 + 0.1^2 + 0.2^2 + 0.1^2) = 0.1$

$MSE = \frac{SSE}{N} = \frac{0.1}{4} = 0.025$

$R^2 = 1 - \frac{0.1}{(3^2 + 1^2 + 1^2 + 3^2)} = 1 - \frac{0.1}{20} = 0.995$

Statistical inference on SLR parameters

Assumption in SLR: Linearity (the dependent variable is a linear function of the independent variable).

$$y_i = m_0 x_i + c_0 + \varepsilon_i$$

we set $\frac{\partial MSE}{\partial m}$ and $\frac{\partial MSE}{\partial c} = 0$, we get

$$\sum x_i(y_i - mx_i - c) = 0, \qquad \sum(y_i - mx_i - c) = 0$$

Residual error, $\varepsilon_i = y_i - \hat{y_i} = y_i - mx_i - c$

- $\sum \varepsilon_i = 0 \Rightarrow E[\varepsilon_i] = 0$
- $\sum y_i = \sum \hat{y_i}$
- $\sum x_i \cdot \varepsilon_i = 0$
- $\sum y_i \varepsilon_i = 0$

$$y_i = mx_i + c$$
$$\Rightarrow x_i = \frac{y_i - c}{m}$$
$$\Rightarrow \sum \frac{y_i - c}{m} \varepsilon_i = 0$$
$$\Rightarrow \sum y_i \varepsilon_i = 0$$

- The simple linear regression line always passes through $(\bar{x}, \bar{y})$.

$$y_1 = mx_1 + c$$
$$\vdots$$
$$y_n = mx_n + c$$
$$\sum y_i = m \sum x_i + c$$
$$\frac{\sum y_i}{n} = m \frac{\sum x_i}{n} + c$$

- The parameters in SLR are unbiased.

Proof:

Assumptions:

- $y_i = m_0 x_i + c_0 + \varepsilon_i$
- $\varepsilon_i \sim N(0, \sigma^2)$, $\sum \varepsilon_i = 0$

To prove:

$E[m] = m_0$ $\quad$ $E[c] = c_0$

$$m = \frac{N\sum x_i y_i - \sum x_i \sum y_i}{N\sum x_i^2 - (\sum x_i)^2} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$$\Rightarrow m = \frac{\sum(x_i - \bar{x})(m_0 x_i + c_0 + \varepsilon_i - m_0 \bar{x} - c_0 - \bar{\varepsilon})}{\sum(x_i - \bar{x})^2}$$

$$= \frac{\sum(x_i - \bar{x}) m_0 (x_i - \bar{x})}{\sum(x_i - \bar{x})^2} + \frac{\sum(x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon})}{\sum(x_i - \bar{x})^2}$$

$$m = m_0 + \frac{\sum(x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon})}{\sum(x_i - \bar{x})^2}$$

$$E[m] = E[m_0] + E\left[\frac{\sum(x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon})}{\sum(x_i - \bar{x})^2}\right] \to 0 \text{ as } \sum x_i \varepsilon_i = 0,\ \sum \varepsilon_i = 0,\ E[\varepsilon] = 0$$

$$E[m] = m_0$$

For c:

$$c = \frac{\sum y_i - m \sum x_i}{N}$$

$$\Rightarrow c = \bar{y} - m\bar{x}$$

$$E[c] = E[\bar{y} - m\bar{x}]$$
$$= E[m\bar{x} + c_0 + \bar{\varepsilon} - m_0 \bar{x}]$$
$$= m_0 \bar{x} + c_0 - m_0 \bar{x}$$
$$= c_0$$

- $E[\hat{y_i}] = E[y_i]$

Proof:

$$E[\hat{y_i}] = E[mx_i + c]$$
$$= E[m_0 x_i + c_0]$$
$$= E[y_i]$$

Confidence intervals for m and c in SLR

mean known, variance known → t-distribⁿ.

$$\theta \in \hat{\theta} \pm t_{\frac{\alpha}{2}, n-2} \cdot SE_{\hat{\theta}}$$

$$\hat{m} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$m \in \left[\hat{m} \pm t_{\frac{\alpha}{2}, n-2} \, SE(\hat{m})\right]$$

↳ standard error of the slope

$$SE_{\hat{m}} = \frac{\sigma_{\hat{m}}}{\sqrt{\sum (x_i - \bar{x})^2}}$$

Ex: Let us assume we have calculated slope $\hat{m} = 2.5$ and intercept $\hat{c} = 5$ for a SLR model with standard errors $SE(\hat{m}) = 0.5$ and $SE(\hat{c}) = 1.2$ for a 95% confidence interval. The critical value from t-distribution for 95% confidence and $n-2$ d.o.f. , $t_{0.025, n-2} = 2.086$. Calculate the 95% C.I for m&c.

95% CI for m :-

$$m \in \left[\hat{m} \pm t_{\alpha/2, n-2} \cdot SE(\hat{m})\right]$$

$$= [2.5 \pm 2.086 \times 0.5]$$

$$= [2.5 \pm 1.043] = (1.457, 3.543)$$

95% CI. for c :-

$$c \in [\hat{c} \pm t_{\alpha/2, n-2} \cdot SE(\hat{c})]$$

$$= [5 \pm 2.086 \times 1.2]$$

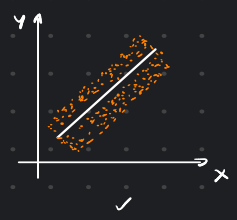
$$= [5 \pm 2.503] = \boxed{(2.497, 7.503)}$$

Assumptions in simple linear regression

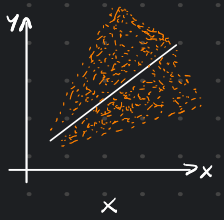
1. Linearity:
There is a linear relationship between X and Y.

2. Independence of errors:
The residual errors (ε_i) are independent of each other. There should be no correlation between errors for different observations.

3. Homoscedasticity (constant variance of errors):
The residuals should have a constant variance across all levels of the independent variable x.



✓



✗

4. Normality of residuals:
The residuals should be normally distributed. This is esp. imp. for confidence intervals and hypothesis testing on parameters.

Multiple linear regression

MLR is an extension of SLR, where instead of a single independent variable (x), we have multiple independent variables (x_1, x_2, ..., x_n) to predict y.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \varepsilon$$

y - Dependent variable
$x_1, x_2, \ldots, x_p$ - Independent variables
β_0 - Intercept
ε - error term (residual)

e.g.

Experience (in years) x_1	Education (in years) x_2	Salary y
1	2	40,000
2	3	50,000
3	3	55,000
4	4	65,000
5	5	75,000

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Assumptions in MLR:
- Linearity: The relationship betn the dependent and independent variables is linear.
- Independence of errors: The residuals (errors) must be independent of each other.
- Homoscedasticity: The variance of residuals should be constant across all levels of independent variables.
- Normality of residuals: The residuals should be normally distributed.
- No multicollinearity: The independent variables should NOT be highly correlated with each other. Variation inflation factor (VIF) is often used to check multicollinearity.

Least squares method

Goal: Find the optimal values of $\beta_1, \beta_2, \ldots, \beta_p$.

Cost function:

$$\underset{(SSE)}{RSS} = \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{N} \left(y_i - (\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n)\right)^2$$

Matrix representation:

$$y = x\beta + \varepsilon$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}_{N\times 1} = \begin{bmatrix} 1 & x_{11} & x_{21} & \cdots & x_{p1} \\ 1 & x_{12} & x_{22} & \cdots & x_{p2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1N} & x_{2N} & \cdots & x_{pN} \end{bmatrix}_{N\times(p+1)} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}_{(p+1)\times 1} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_N \end{bmatrix}_{N\times 1}$$

$$y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{22} + \cdots + \beta_p x_{p1} + \varepsilon_1$$
$$\vdots$$
$$y_N = \beta_0 + \beta_1 x_{1N} + \cdots + \beta_N x_{pN} + \varepsilon_N$$

$$RSS = \sum_{i=1}^{N} \left(y_i - (\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_p x_{pi})\right)^2$$
$$= (y - x\beta)^T (y - x\beta)$$

$$y - x\beta = \begin{bmatrix} y_1 - (\beta_0 + \beta_1 x_{11} + \cdots + \beta_p x_{p1}) \\ y_2 - (\beta_0 + \beta_1 x_{12} + \cdots + \beta_p x_{p2}) \\ \vdots \\ y_n - (\beta_0 + \beta_1 x_{1N} + \cdots + \beta_p x_{pN}) \end{bmatrix}_{n\times 1}$$

$$(y - x\beta)^T (y - x\beta) = \begin{bmatrix} RSS \end{bmatrix}_{1\times 1}$$

Claim: optimal values of β which minimize RSS: $\beta = (x^T x)^{-1} x^T y$

Proof:

$$RSS = (y - x\beta)^T (y - x\beta)$$

$$= (y^T - \beta^T x^T)(y - x\beta)$$

$$= y^T y - y^T x\beta - \beta^T x^T y + \beta^T x^T x \beta$$

$$= y^T y - y^T x \beta - (y^T x \beta)^T + \beta^T X^T X \beta$$

$\frac{\partial}{\partial x} x^T A x = 2Ax$

transpose of 1x1 matrix is same as that matrix

$$= y^T y - y^T x\beta - y^T x\beta + \beta^T x^T x \beta$$

$$= y^T y - 2y^T x\beta + \beta^T x^T x \beta$$

$$\frac{\partial RSS}{\partial \beta} = -2y^T x + 2x^T x \beta$$

$1 \times (p+1)$

$$x^T x \beta = y^T x$$

$$\beta = (x^T x)^{-1} y^T x$$

Dataset :-

x_1	x_2	y
1	1	1
2	3	7
3	4	9
4	6	13

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 6 \end{bmatrix} \quad y = \begin{bmatrix} 1 \\ 7 \\ 9 \\ 13 \end{bmatrix}$$

$$\beta = (X^T X)^{-1} X^T y$$

$$\rightarrow X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 4 & 6 \end{bmatrix}_{3\times4} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 6 \end{bmatrix}_{4\times3} = \begin{bmatrix} 4 & 10 & 14 \\ 10 & 30 & 43 \\ 14 & 43 & 62 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 11/4 & -9/2 & 5/2 \\ -9/2 & 13 & -8 \\ 5/2 & -8 & 5 \end{bmatrix}$$

$$(X^T X)^{-1} X^T = \begin{bmatrix} 11/4 & -9/2 & 5/2 \\ -9/2 & 13 & -8 \\ 5/2 & -8 & 5 \end{bmatrix}_{3\times3} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 4 & 6 \end{bmatrix}_{3\times4} = \begin{bmatrix} 3/4 & 5/4 & -3/4 & -1/4 \\ 1/2 & -5/2 & 5/2 & -1/2 \\ -1/2 & 3/2 & -3/2 & 1/2 \end{bmatrix}$$

$$(X^T X)^{-1} X^T y = \begin{bmatrix} 3/4 & 5/4 & -3/4 & -1/4 \\ 1/2 & -5/2 & 5/2 & -1/2 \\ -1/2 & 3/2 & -3/2 & 1/2 \end{bmatrix}_{3\times4} \begin{bmatrix} 1 \\ 7 \\ 9 \\ 13 \end{bmatrix}_{4\times1} = \begin{bmatrix} -1/2 \\ -1 \\ 3 \end{bmatrix}_{3\times1}$$

$$\boxed{y = \frac{-1}{2} - x_1 + 3x_2}$$

x_1	x_2	y	$\hat{y}$	$\varepsilon = y - \hat{y}$	ε^2
1	1	1	1.5	-0.5	0.25
2	3	7	6.5	0.5	0.25
3	4	9	8.5	0.5	0.25
4	6	13	13.5	-0.5	0.25
					1

$$MAE = \frac{1}{4} \cdot 2 = \frac{1}{2}$$

$$SSE = 1$$

$$MSE = 1/4$$

$$RMSE = \sqrt{\frac{1}{4}} = 1/2$$

$$MAPE = \frac{1}{N} \sum \frac{|y_i - \hat{y_i}|}{y_i} \times 100 = \frac{1}{4} \times \left(\frac{0.5}{1} + \frac{0.5}{7} + \frac{0.5}{9} + \frac{0.5}{13} \right) \times 100 = 16.63\ \%$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y_i})^2}{\sum (y_i - \bar{y})^2} = 0.987$$

$$R^2_{adj} = 1 - \frac{\sum (y_i - \hat{y_i})^2 / (N-k-1)}{\sum (y_i - \bar{y})^2 / (N-1)} = 0.96 \quad \rightarrow \{k=2, N=4\}$$

Multicollinearity in MLR

Multicollinearity occurs when 2/more independent vars are highly correlated with each other. In other words, one predictor can be linearly predicted from the others with a substantial degree of accuracy.

Perfect collinearity example: $x_1 = 2x_2 + 3x_3 + 5$.

Imperfect collinearity example: 2 or more vars are highly correlated but relationship is not perfect. ← more common

Detecting multicollinearity:

1. Correlation matrix:

A high correlation betn 2 independent variables suggests potential multicollinearity.

2. Variance inflation factor:

To check for x_i in $\{x_1, x_2, \ldots, x_n\}$, fit $x_i = \beta_0 + \beta_1 x_2 + \beta_2 x_3 + \ldots$ then find R^2.

VIF quantifies how much variance of a coefficient is inflated due to multicollinearity.

$$VIF(x_i) = \frac{1}{1 - R_i^2}$$

If R^2 goes close to 1, VIF will go up.

$R_i^2 \uparrow$ VIF ↑↑↑

VIF = 1 → zero correlation betn predictor and other variables

VIF ∈ (5,10) → moderate correlation

VIF > 10 → high correlation

Steps to calculate R_i^2:

$\{x_1, x_2, x_3\}$

① Fit the model $x_1 = \beta_0 + \beta_1 x_2 + \beta_2 x_3 + \varepsilon$

② Calculate R_i^2.

③ $VIF = \frac{1}{1 - R^2}$

Example:

```
data = {'X1': [1, 2, 3, 4],
        'X2': [2, 7, 4, 5.1],
        'X3': [2.1, -4.1, 4.2, 5]}
```

```
R_1^2 for X1: 0.9999662932628757
VIF for X1: 29667.659504136267
```

Gradient descent for MLR

$$w_j = w_j - \alpha \cdot \frac{\partial J(w)}{\partial w_j}$$

For MLR, $J(\beta) = \frac{1}{N} \sum_{i=1}^{N} \left(y_i - (\beta_0 + \beta_1 x_{1i} + \ldots + \beta_p x_{pi})\right)^2$ — MSE for MLR

$$\frac{\partial J(\vec{\beta})}{\partial \beta_j} = \frac{-2}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)\, x_{ji} \qquad (\text{for } j=0,\ x_{ji} = 1)$$

$$\beta_j = \beta_j - \alpha \frac{\partial J(\vec{\beta})}{\partial \beta_j}$$

Steps:

1. Initialize the coefficients: $\beta_0, \beta_1, \ldots, \beta_p$.
2. Calculate the predicted values using the current coefficients: $\hat{y} = \beta x$
3. Compute the cost function J(w). Calculate MSE.
4. Update:

$$\beta = \beta - \alpha \frac{\partial J(\beta)}{\partial \beta}$$

Repeat until convergence.

Polynomial regression

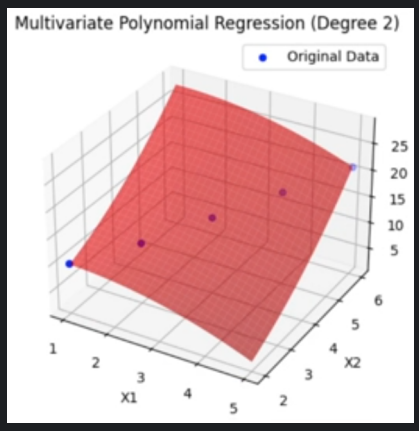
simple polynomial regression : $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \ldots + \beta_n x^n$ — $n \rightarrow$ degree of polynomial

multiple polynomial regression : consider 2 vars: $\{x_1, x_2\}$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2 + \beta_5 x_1 x_2$$

→ degree of polynomial → 2
of predictors → 2
of parameters → 6

If we keep increasing the degree for data that isn't really that complex, we'd just overfit.



Regularization

A problem in regression is overfitting. The # of predictors will usually be high. The # of training examples may be low.
We don't penalize the # of parameters in regression.
A way to prevent overfitting is to penalize the # of parameters (regularization).

Regularization is a technique to prevent overfitting in ML models, especially in regression models like MLR.
Overfitting happens when a model is too complex and captures not just the underlying pattern, but also noise in the data, leading to poor performance on unseen data.
Regularization adds a penalty term to the loss function which helps in reducing the complexity of the model by shrinking the magnitude of the coefficients.
This results in a simpler, generalizable model.

$$\text{cost} = SSE + \lambda \sum_{i=0}^{p} |\beta_i|^q$$

← this term penalizes high β values, so they'll be restricted.
Then some β values will get reduced.
so some parameters will be given less importance.
The low importance parameters will get low β values.

least absolute shrinkage and selection operator

Lasso regression (L1 regularization)

$$\text{cost} = \text{SSE} + \lambda \underbrace{\sum_{i=1}^{p} |\beta_i|}_{\text{L1 penalty}}$$

$\lambda \rightarrow$ tuning parameter

If $\lambda = 0$, Lasso becomes OLS (original least squares).

when λ increases, some coefficients shrink to zero, making the model simpler and selecting only the important features.

Lasso performs both variable selection and regularizaⁿ.

It is useful when there are many features, but only a few of them are significant.

How does it shrink some coefficients to zero?

$$\text{cost} = E(\theta) + \lambda R(\theta)$$

as $\lambda \uparrow$: $E(\theta) \uparrow$, $R(\theta) \downarrow$

Say $\lambda_1 > \lambda_2$. For $\lambda_1 \rightarrow \theta_1$ is optimal.
For $\lambda_2 \rightarrow \theta_2$ is optimal.

$$E(\theta_1) + \lambda_1 R(\theta_1) \leq E(\theta_2) + \lambda_1 R(\theta_2)$$
$$E(\theta_2) + \lambda_2 R(\theta_2) \leq E(\theta_1) + \lambda_2 R(\theta_2)$$

$$\lambda_2 [R(\theta_2) - R(\theta_1)] \leq E(\theta_1) - E(\theta_2) \leq \lambda_1 [R(\theta_2) - R(\theta_1)]$$

$R(\theta_2) - R(\theta_1) > 0$

$R(\theta_2) > R(\theta_1)$ As $\lambda \downarrow$ $R(\theta) \uparrow$

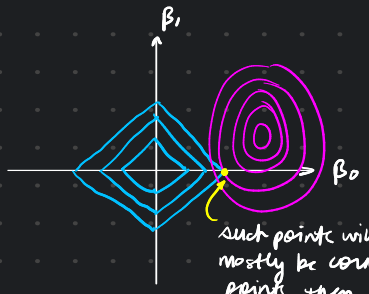
$E(\theta_1) - E(\theta_2) > 0$

$E(\theta_1) > E(\theta_2)$ As $\lambda \downarrow$, $E(\theta) \downarrow$

} As $\lambda \uparrow$, $R(\theta) \downarrow$, $E(\theta) \uparrow$

Hence, increasing λ causes increasing $E(\theta)$, decreasing $R(\theta)$.

Contours: set of points for which the function value is the same.



such points will mostly be corner points, then one of the coefficients is zero.
In more dimension, a greater # of coefficients will be zero.

$R(\theta) = |\beta_0| + |\beta_1|$ for 2 parameters

$E(\theta) = \text{SSE}$

we want min of $R(\theta) + E(\theta)$.
We'll get that at a point where the 2 contours are tangential

Lasso creates sparsity, making some coeffs zero. This is feature selection.
We cannot apply least squares to lasso, as it isn't differentiable.
We apply other approximation techniques such as coordinate descent.

↳ It solves for 1 coeff. at a time, keeping the others fixed, and iterates over all coeffs until convergence.

Ridge regression (L2 regularization)

$$cost = SSE + \lambda \sum_{i=1}^{p} \beta_i^2$$

$$J(\beta) = (y - x\beta)^T (y - x\beta) + \lambda \beta^T \beta$$

$$= y^T y - 2y^T x\beta + \beta^T x^T x \beta + \lambda \beta^T \beta$$

$$\frac{\partial J(\beta)}{\partial \beta} = -2x^T y + 2x^T x \beta + 2\lambda\beta$$

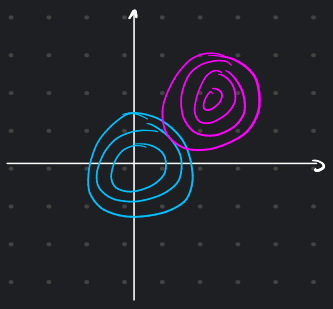
$$\frac{\partial J(\beta)}{\partial \beta} = 0$$

$$\Rightarrow -2x^T y + 2x^T x \beta + 2\lambda\beta = 0$$

$$\Rightarrow x^T x \beta + \lambda\beta = x^T y$$

$$\Rightarrow (x^T x + \lambda I)\beta = x^T y$$

$$\Rightarrow \beta = (x^T x + \lambda I)^{-1} x^T y$$



Ridge regression is not for feature selection.
It distributes the penalty evenly among all the coefficients. It doesn't let one of them grow.

Elastic net regularization

$$cost = SSE + \alpha \left[l_1\text{-ratio} \cdot \sum_{i=0}^{p} |\beta_i| + (1 - l_1\text{-ratio}) \cdot \sum_{i=0}^{p} \beta_i^2 \right]$$